

CHINTAN VAGHAMSHI

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EDUCATION

Indian Institute of Technology Jodhpur

B.Tech in Electrical Engineering | CGPA: 7.50/10.00

Aug 2024 – Jul 2028

Jodhpur, Rajasthan

Ashadeep Vidhyalay

Class XII | 87%

2024

Surat, Gujarat

Ashadeep Vidhyalay

Class X | 95%

2022

Surat, Gujarat

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Technologies: Node.js, React, Next.js, Express, MongoDB, Redis, BullMQ, Socket.io, REST APIs, WebSockets, JWT, OAuth 2.0, WebCrypto API

Tools: Git, GitHub Actions, Nginx, Vercel, Render, Cloudflare R2, Cloudinary, Postman, VS Code

EXPERIENCE

Open Source Contributor | NetworkX

Nov 2025 – Jan 2026

Python, Graph Theory, Open Source

Remote

- Contributed enhancements to the NetworkX graph-processing codebase, focusing on algorithm correctness and edge-case handling across graph traversal APIs.
- Improved contributor-facing documentation for core graph APIs — changes reviewed and merged into the main branch.

PROJECTS

SecureShare | [GitHub](#) | [Live](#) | JavaScript, React, Node.js, Express, MongoDB, Cloudinary, WebCrypto API

- Architected a **zero-knowledge E2E encrypted** file sharing platform across **18 REST APIs** — browser encrypts files via **WebCrypto (AES-256-GCM + RSA-2048 OAEP)** before upload; **server mathematically incapable of reading user files** even under full infrastructure compromise.
- Implemented **PBKDF2 key derivation** (100K iterations), MongoDB-persisted account lockout, **JWT refresh token rotation** with HttpOnly cookies, and **SHA-256 integrity verification** — defeating **4 simultaneous attack vectors** (XSS, CSRF, brute force, tampering).
- Built complete security audit trail across **8 action types** with user dashboard surfacing real-time failed login attempts and **INTEGRITY_FAIL** alerts — **zero plaintext file exposure** verified across MongoDB, Cloudinary, and server RAM.

WayPoint | [GitHub](#) | JavaScript, React, Node.js, Express, MongoDB, Redis, BullMQ, Socket.io, Cloudflare R2

- Architected a **real-time collaborative project management platform** (MERN) with **Socket.io** rooms — live kanban board, drag-and-drop, presence indicators, and notifications with sub-100ms WebSocket latency.
- Engineered **JWT auth** with access/refresh token rotation, **Google OAuth 2.0** via Passport.js, and **4-tier RBAC** (owner/admin/member/viewer) across **15+ REST endpoints** with **Zod** schema validation.
- Integrated **Cloudflare R2** via **presigned URLs** for zero-server-touch file uploads (25MB attachments); implemented **Redis (BullMQ)** background job queues for async task processing and notification delivery.

Hybrid Repo AI Agent | [GitHub](#) | Python, Ollama, OpenRouter, Sentence Transformers, FAISS

- Built a **repository-aware AI coding agent** combining **semantic retrieval** (embedding indexing + reranking) and **dependency graph traversal** for multi-file codebase understanding — persistent memory enables cross-session context.
- Designed a **hybrid local/cloud LLM inference pipeline** routing requests between **Ollama local models** and **OpenRouter cloud LLMs** based on task complexity heuristics — reduces inference latency and API cost.
- Implemented **DFS-based context expansion** and multi-file patch generation for repository-level reasoning — agent resolves cross-file dependencies during code editing workflows.

COURSEWORK

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|--------------------------------|------------------------|---|
| • Data Structures & Algorithms | • Pattern Recognition | • Probability & Statistics and Stochastic Processes |
| • Machine Learning | • Discrete Mathematics | • Operating Systems (ongoing) |

ACHIEVEMENTS & PARTICIPATION

- Solved **300+** DSA problems across LeetCode and competitive programming platforms.
- Achieved **Gold Medal** In SOF Science Olympiad
- FFE Scholar** at IIT Jodhpur — merit-based financial excellence award.
- Cleared **2 rounds** of **Smart India Hackathon** (national-level government hackathon).
- Ranked **3rd in State** in Karate; competed at **National Level** twice.

LEADERSHIP

- Core Member**, E-Cell, IIT Jodhpur — organized and managed entrepreneurship events over 9 months.
- Core Member**, Programming Club, IIT Jodhpur — co-organized competitive programming contests and workshops.